$[0, 2\pi]$

$5\sec x = 4 + 3\sec x$

$-3\sec x \quad -3\sec x$

$2\sec x = 4$

$\sec x = \frac{4}{2}$

$\sec x = \frac{2}{1}$

$\cos x = \frac{1}{2}$

$x = \left(\frac{\pi}{3}, \frac{5\pi}{3}\right)$

$\sqrt{3}\tan x - 1 = 0$

$\sqrt{3}\tan x = 1$

$\tan x = \frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$

$x = \frac{\pi}{6} + n\pi$

$2\sin 3x = \sqrt{3}$

$\sin 3x = \frac{\sqrt{3}}{2}$

$3x = \frac{\pi}{3}, \frac{2\pi}{3}$

$x = \frac{\pi}{9} + \frac{2n\pi}{3}$ & $\frac{2\pi}{9} + \frac{2n\pi}{3}$

$x = \frac{\pi}{9}, \frac{7\pi}{9}, \frac{13\pi}{9}$

$\cos\left(x + \frac{\pi}{6}\right) = \frac{-\sqrt{3}}{2}$

$x + \frac{\pi}{6} = \frac{5\pi}{6}$ & $\frac{7\pi}{6}$

$x + \frac{\pi}{6} = \frac{5\pi}{6} + 2n\pi$

$x = \frac{5\pi}{6} + 2n\pi - \frac{\pi}{6}$

$x = \frac{4\pi}{6} + 2n\pi$

$x = \frac{2\pi}{3} + 2n\pi$

$2\sin^2 x - \sin x - 1 = 0$

$2v^2 - v - 1 = 0$

$(2v + 1)(v - 1)$

$2v + 1 = 0 \qquad v - 1 = 0$

$2v = -1 \qquad v = 1$

$v = \frac{-1}{2}$

$\sin x = \frac{1}{2} \qquad \sin x = 1$

$\sin x = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{\pi}{2}$

$2\cos^2 x + 3\cos x + 1 = 0$

$2v^2 + 3v + 1 = 0$

$(2v + 1)(v + 1)$

$2\sin x + 1 = 0 \qquad \sin x + 1 = 0$

$2\sin x = -1 \qquad \sin x = -1$

$\sin x = \frac{-1}{2}$

$x = \frac{7\pi}{6}, \frac{11\pi}{6} \qquad x = \frac{3\pi}{2}$

COS

$\cos 2x = \cos x$

$2\cos^2 x - 1 = \cos x$

$2\cos^2 x - \cos x - 1 = 0$

$2v^2 - v - 1$

$(2v + 1)(v - 1)$

$2\cos x + 1 = 0 \qquad \cos x - 1 =$

$2\cos x = -1 \qquad \cos x = 1$

$\cos x = \frac{-1}{2}$

$3\cos x - 1 = 0$

$3\cos x = 1$

$\cos x = \frac{1}{3}$

$x = \cos^{-1}\left(\frac{1}{3}\right)$

$[0, 2\pi]$

$2\tan x + 1 = \tan x$

$\tan x + 1 = 0$

$\tan x = -1$

$x = \left(\frac{3\pi}{4}, \frac{7\pi}{4}\right)$

$2\sin x + 3 = 2$

$2\sin x = -1$

$\sin x = \frac{-1}{2}$

$x = \left(\frac{7\pi}{6}, \frac{11\pi}{6}\right)$

$x = \left(\frac{7\pi}{6} + 2n\pi\right)$ & $\left(\frac{11\pi}{6} + 2n\pi\right)$

$2\cos(2x) - 1 = 0$

$2\cos(2x) = 1$

$\cos(2x) = \frac{1}{2}$

$2x = \frac{\pi}{3}$ & $\frac{5\pi}{3}$

$x = \frac{\frac{\pi}{3} + 2n\pi}{2}$ & $\frac{\frac{5\pi}{3} + 2n\pi}{2}$

$x = \frac{\pi}{6} + n\pi$ & $\frac{5\pi}{6} + n\pi$

$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

$x + \frac{\pi}{6} = \frac{7\pi}{6} + 2n\pi$

$x = \pi + 2n\pi$

$x = \left(\frac{2\pi}{3}, \pi\right)$

$\tan\left(\frac{x}{2}\right) = 1$

$\frac{x}{2} = \frac{\pi}{4}$, ~~$\frac{5\pi}{4}$~~

$\frac{x}{2} = \frac{\pi}{4} + n\pi$

$x = \frac{2\pi}{4} + 2n\pi$

$x = \frac{\pi}{2} + 2n\pi$

$x = \frac{\pi}{2}$

$x = \frac{7\pi}{6} + 2n\pi \qquad x = \frac{11\pi}{6} + 2n\pi$

$x = \frac{\pi}{2} + 2n\pi$

$x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

$x = \frac{7\pi}{6} + 2n\pi \qquad x = \frac{11\pi}{6} + 2n\pi$

$x = \frac{3\pi}{2} + 2n\pi$

$\sin 2x = \sin x$

$2\sin x \cdot \cos x = \sin x$

$-\sin x$

$2\sin x \cdot \cos x - \sin x = 0$

$\sin(2 \cdot \cos x - 1)$

$\sin x = 0 \qquad 2\cos x - 1 = 0$

$x = 0, \pi \qquad 2\cos x = 1$

$\cos x = \frac{1}{2}$

$x = \frac{\pi}{3}, \frac{5\pi}{3}$

$x = \frac{2\pi}{3}, \frac{4\pi}{3}, 0$